Searching

Programming · Level 3 · Lesson 3



In this lesson you will

- Write linear search.
- Explain binary search, and why the list must be sorted.
- Count the worst-case looks for 1,000 items.

Key ideas

- Linear: one by one, any list, up to n looks.
- Binary: halve a sorted list, about 10 looks for 1,000.
- Return -1 only after checking everything.

Activity: Find the card

1. Warm-up (3 min): Find a number in 16 face-down cards.
2. Sorted (7 min): Now the cards are in order. Halve!
3. Count (7 min): Record the looks each way.
4. Explain (3 min): Why must the cards be sorted?

Playing cards or numbered paper squares both work. No phone is needed until the code.

Let's discuss

1. Why does binary search need a sorted list?
2. When is linear search good enough?
3. Why does a list a thousand times bigger need only about 10 more looks?

Exit ticket

- How many looks for 1,000 items: linear worst case, and binary worst case?